

CS120B Custom Laboratory Project Proposal

Intro:

The snake grows longer by eating the food that appears randomly and the game ends when the snake hits the wall or itself. The goal of the game is to get the longest snake which will probably fill up the entire lcd screen

Hardware:

AT mega328 Microcontroller

Joystick for the input to control the snake

The button to turn on and off the game

The reset button for restarting the game

Basic Functionality:

The game start by user pressing the start button and then 1 pixel snake will appear and depends on which way you move the joystick, it will start moving that direction and the joystick can move left or right or up and down, then the food will appear on the screen and when the snake eats the food the snake will grow in a pixel, the game ends when the screen is full of snake and it hit a wall or just hitting itself or hitting a wall when the snake is not fully yet. When you eat the food the passive buzzer makes a sound and when the game is over it will also output a game over song.

Complexities

- 1.HiLetgo 1.8" inch ST7735R SPI 128 * 160 TFT LCD Display Module with PCB for displaying the main game area
- 2.Passive buzzer is used for the sound effects of eating the food or game over
- 3.Use the ATmega328's EEPROM to save the high score (1/2 complexity) and use the LCD from the lab kit to display the score score (1/2 complexity).